

By: AKH  
 Rev.--- By: ---  
 Chk: ---

Date Jun 2026  
 Date ---  
 Date ---

MacTest

### Macaulay Beam Analysis - Example Worksheet

#### 103 - (2) EQ Span - Point UDL

Beam Sect = W14x61 W14x90  
 A = 17.90 26.50 in<sup>2</sup>  
 I = 640.00 999.00 in<sup>4</sup>

E = 29,000,000 lb / in<sup>2</sup>

Length Units in  
 Force Units lb

#### Beam Segments (First Segment Begins at 0)

|                                    | Segment No. | X End  | A               | I               | E                    |
|------------------------------------|-------------|--------|-----------------|-----------------|----------------------|
|                                    |             | in     | in <sup>2</sup> | in <sup>4</sup> | lb / in <sup>2</sup> |
| DO NOT<br>DELETE ROWS<br>(GRAPHED) | 1           | 600.00 | 17.90           | 640.00          | 2.900E+07            |
|                                    | 2           |        |                 |                 |                      |
|                                    | 3           |        |                 |                 |                      |
|                                    | 4           |        |                 |                 |                      |
|                                    | 5           |        |                 |                 |                      |

#### Supports

|                                    | Supp. No. | X     | Type | Imposed Movement |             |
|------------------------------------|-----------|-------|------|------------------|-------------|
|                                    |           | in    | ---  | * dY (vert)      | ** dr (rot) |
| DO NOT<br>DELETE ROWS<br>(GRAPHED) | 1         | 0.0   | P    | 0.0              | 0.000       |
|                                    | 2         | 300.0 | P    | 0.0              | 0.000       |
|                                    | 3         | 600.0 | P    | 0.0              | 0.000       |
|                                    | 4         |       |      |                  |             |
|                                    | 5         |       |      |                  |             |

\* (settlement) \*\* (Not Working)

#### Distributed Loads (downward loads entered as NEGATIVE) - UNFACTORED

|                                    | Load No. | x1  | x2    | w1 (at x1) | w2 (at x2) |
|------------------------------------|----------|-----|-------|------------|------------|
|                                    |          | in  | in    | lb / in    | lb / in    |
| DO NOT<br>DELETE ROWS<br>(GRAPHED) | 1        | 0.0 | 600.0 | -5.1       | -5.1       |
|                                    | 2        | 0.0 | 0.0   | 0.0        | 0.0        |
|                                    | 3        | 0.0 | 0.0   | 0.0        | 0.0        |
|                                    | 4        | 0.0 | 0.0   | 0.0        | 0.0        |
|                                    | 5        | 0.0 | 0.0   | 0.0        | 0.0        |

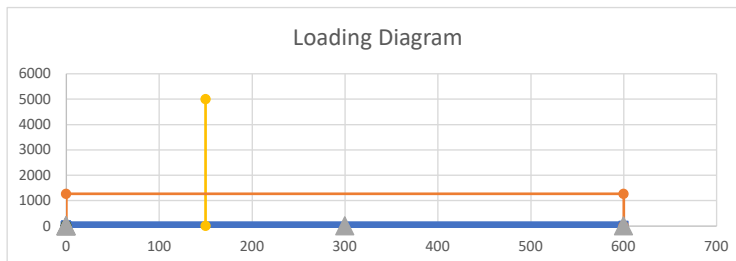
| Load    |
|---------|
| lb / ft |
| 61.0    |
| 0.0     |
| 0.0     |
| 0.0     |

#### Point Loads - UNFACTORED

|                                    | Load No. | X   | Py       | My    |
|------------------------------------|----------|-----|----------|-------|
|                                    |          | in  | lb       | lb-in |
| DO NOT<br>DELETE ROWS<br>(GRAPHED) | 1        | 150 | -5000.00 | 0.00  |
|                                    | 2        | 0   | 0.00     | 0.00  |
|                                    | 3        |     |          |       |
|                                    | 4        |     |          |       |
|                                    | 5        |     |          |       |

| My Load |
|---------|
| lb-ft   |
| 0.0     |
| 0.0     |
| 0.0     |

Loading Diagram



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## BEAM REACTIONS

| Position (X) | Enforced Disp (dy) | Enforced Rot (theta) | Vert Reac (Ry) | Internal Moment (Mz) | Type  |
|--------------|--------------------|----------------------|----------------|----------------------|-------|
| (L)          | (L)                | (rad)                | (F)            | (F-L)                | (---) |
| 0.00         | 0.0                | 0.0                  | 2,603.1        | 0.0                  | P     |
| 300.00       | 0.00               | 0.00                 | 5,343.7        | 0.0                  | P     |
| 600.00       | 0.00               | 0.00                 | 103.1          | 0.0                  | P     |

| Vert Reac |
|-----------|
| (kip)     |
| 2.603     |
| 5.344     |

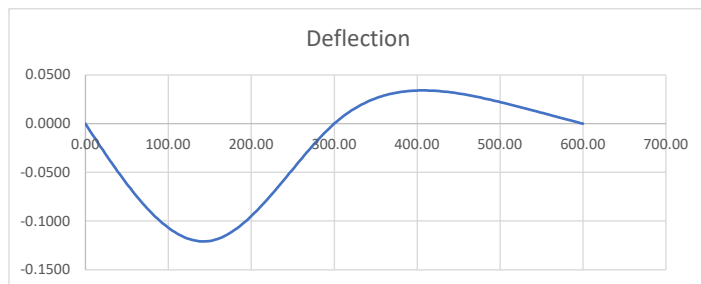
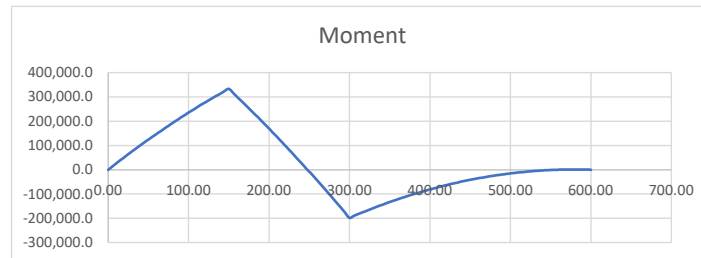
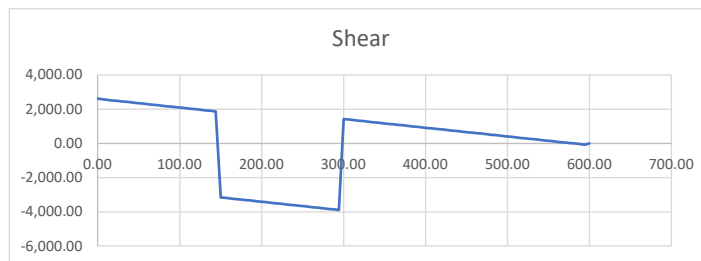
## BEAM REACTIONS EQUILIBRIUM CHECK

| Item        | Units | Loads       | Reactions  | Residual | Check Sum |
|-------------|-------|-------------|------------|----------|-----------|
| Sum Fy      | F     | -8,050.0    | 8,050.0    | 0.0      | OK        |
| Sum M @ x=0 | F-L   | -1.6650E+06 | 1.6650E+06 | 0.0      | OK        |

## BEAM MAX/MIN LOAD EFFECTS

| Item     | Position (X) | Shear (V) | Moment (M) | Defl. (delta) |
|----------|--------------|-----------|------------|---------------|
| (---)    | (L)          | (F)       | (F-L)      | (L)           |
| Max +V   | 0.00         | 2,603.1   | 0.0        | 0.0000        |
| Max -V   | 294.00       | -3,891.4  | -174,369.7 | -0.0047       |
| Max +M   | 150.00       | -3,159.4  | 333,282.8  | -0.1204       |
| Max -M   | 300.00       | 1,421.9   | -197,809.4 | 0.0000        |
| Max Defl | 144.00       | 1,871.1   | 322,147.5  | -0.1209       |

| Max/Min Value |
|---------------|
| (---)         |
| 2.60 kip      |
| -3.89 kip     |
| 27.77 kip-ft  |
| -16.48 kip-ft |
| -0.121 in     |



## INPUT SUMMARY

### General Info

|               |        |
|---------------|--------|
| Beam Length:  | 50 ft  |
| Section Name: | W14x61 |
| Self Weight:  | False  |

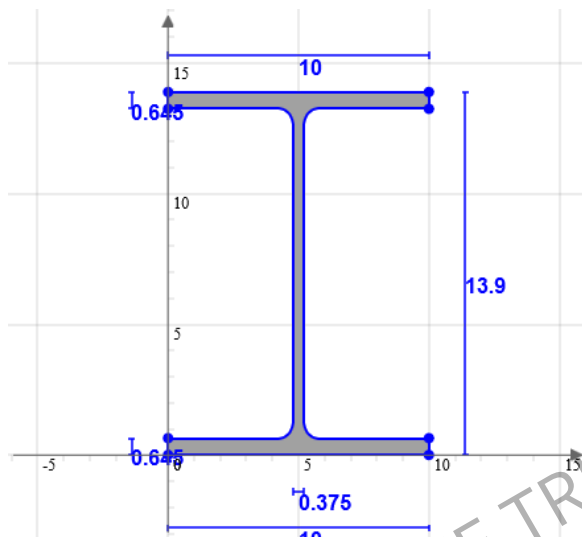
### Supports

| Support Type | Location |
|--------------|----------|
| Pinned       | 0 ft     |
| Roller       | 25 ft    |
| Roller       | 50 ft    |

### Loads

| Load Type        | Location      | Magnitude        | Load Case |
|------------------|---------------|------------------|-----------|
| Point Load       | 12.5 ft       | -5000 lb         | DL        |
| Distributed Load | 0 ft to 50 ft | -61 lb to -61 lb | DL        |

### Beam Section



| Geometric Properties |      |                 |
|----------------------|------|-----------------|
| A                    | 17.9 | in <sup>2</sup> |
| C <sub>z</sub>       | 5    | in              |
| C <sub>y</sub>       | 6.95 | in              |

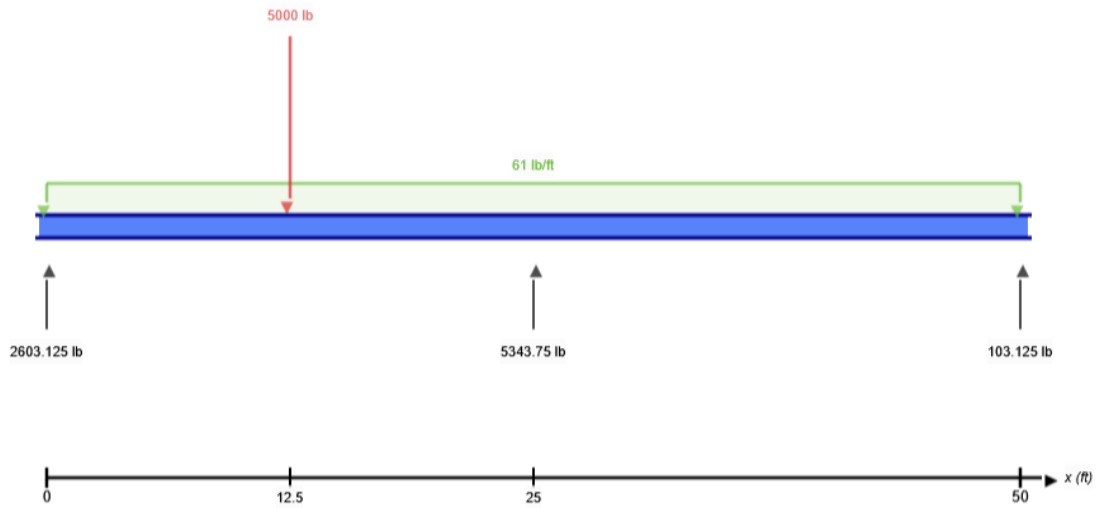
| Bending Properties |     |                 |
|--------------------|-----|-----------------|
| I <sub>z</sub>     | 640 | in <sup>4</sup> |
| I <sub>y</sub>     | 107 | in <sup>4</sup> |

| Shear Properties |        |                 |
|------------------|--------|-----------------|
| A <sub>z</sub>   | 11.277 | in <sup>2</sup> |
| A <sub>y</sub>   | 4.99   | in <sup>2</sup> |

| Torsion Properties |       |                 |
|--------------------|-------|-----------------|
| J                  | 2.19  | in <sup>4</sup> |
| r <sub>r</sub>     | 0.991 | in              |

| Shape  | Material         | E (ksi) | v    | ρ (lb/ft <sup>3</sup> ) |
|--------|------------------|---------|------|-------------------------|
| W14x61 | Structural Steel | 29000   | 0.27 | 490                     |

### FREE BODY DIAGRAM



### RESULT SUMMARY

| Check               | Status | Limit  | Ratio | Max       |
|---------------------|--------|--------|-------|-----------|
| Deflection          | PASS   | L/250  | 0.05  | L/4980    |
| Custom Stress Limit | PASS   | 39 ksi | 0.093 | 3.619 ksi |
| Material Yield      | PASS   | 36 ksi | 0.101 | 3.619 ksi |
| Material Strength   | PASS   | 58 ksi | 0.062 | 3.619 ksi |

## ANALYSIS RESULTS

### Reactions

| Support at | $R_x$ (lb) | $R_y$ (lb) | M (kip-ft) |
|------------|------------|------------|------------|
| 0          | 0          | 2603.125   | 0          |
| 25         | 0          | 5343.75    | 0          |
| 50         | 0          | 103.125    | 0          |

### Force Extremes

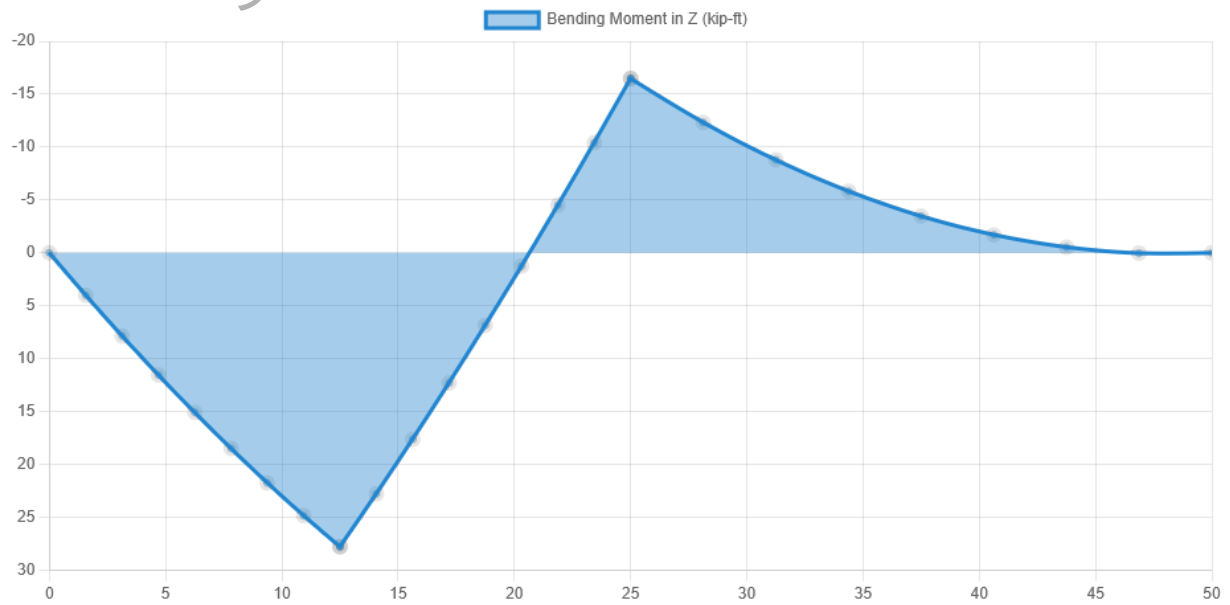
| Result         | Max           | Min            |
|----------------|---------------|----------------|
| Bending Moment | 27.773 kip-ft | -16.484 kip-ft |
| Shear          | 2603.125 lb   | -3921.875 lb   |
| Displacement Y | 0.034 in      | -0.12 in       |

### Stress Extremes

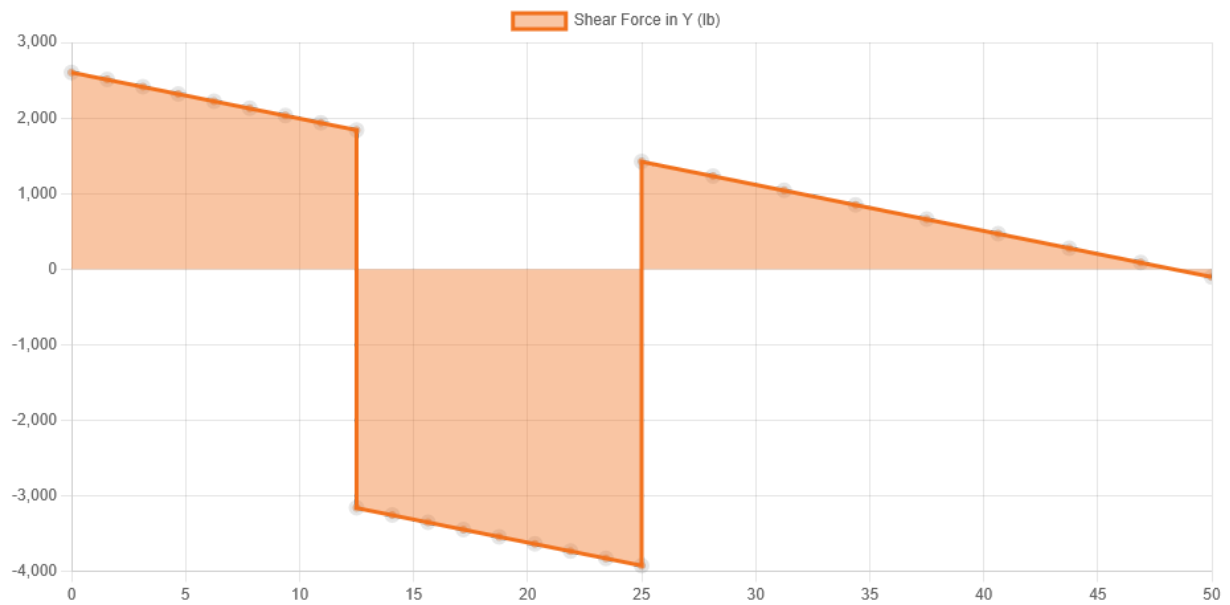
| Result                     | Max (ksi) | Min (ksi) |
|----------------------------|-----------|-----------|
| Bending Stress             | 3.619     | -3.619    |
| Shear Stress Total         | 0.837     | 0.019     |
| Max Combined Normal Stress | 3.619     | 0         |
| Min Combined Normal Stress | 0         | -3.619    |

## DIAGRAMS

### Bending Moment Diagram



### Shear Force Diagram



Displacement



|                                                                                                                                    |     |        |     |         |     |
|------------------------------------------------------------------------------------------------------------------------------------|-----|--------|-----|---------|-----|
| Location (ft)                                                                                                                      | 0   | 12.5   | 25  | 34.375  | 50  |
| Total Deflection (in)                                                                                                              | 0 ? | 0.12 ? | 0 ? | 0.034 ? | 0 ? |
| Deflection/Span                                                                                                                    | -   | L/2490 | -   | L/8785  | -   |
| The Deflection/Span results are calculated using the analysis results and the Deflection Limit of L/250 set in the model settings. |     |        |     |         |     |